

Assignment 6a

Friday, 5 April 2024 1:11 pm

ASSIGNMENT NO. 6(a)

(ROTATION) $\rightarrow$ SPECIAL CASE
(MATRIX) OF
ORTHOGONAL
MATRIX

RECALL THAT $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

IS THE ROTATION MATRIX,

(1) PROVE THAT THE TERMINAL POINT OF $\underline{e_1}$ IN $\underline{R^2}$ REACHES $(\cos \theta, \sin \theta)$ AFTER A ROTATION THROUGH AN ANGLE θ (COUNTERCLOCKWISE)

1. The terminal point $e_1 \in R^2$ is $e_1 = (1, 0)$.

$$\therefore \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

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(2) FROM (1) IF IT REACHES $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ THEN THROUGH WHICH ANGLE ROTATION HAS TAKEN PLACE. ALSO WRITE

TAKEN PLACE. ALSO WRITE DOWN THE ROTATION MATRIX IN THIS CASE.

2. The angle rotation took place is $\theta = \frac{\pi}{4}$.

$$\therefore \begin{bmatrix} \frac{\cos \pi}{4} & -\frac{\sin \pi}{4} \\ \frac{\sin \pi}{4} & \frac{\cos \pi}{4} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

(3) WRITE DOWN THE ROTATION MATRIX WHEN ROTATION IS ABOUT Z-AXIS IN THREE DIMENSIONS.

3. The rotation matrix when rotation is about z-axis in three dimension is,

$$R = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(4) CAN WE SAY THAT THE ROTATION MATRIX IS A TRANSITION MATRIX FROM ONE ORTHONORMAL BASIS TO ANOTHER.

Solution: To make notation simpler, let $\mathcal{S}$ and $\mathcal{T}$ be the bases. Form matrices S and T from the respective basis vectors. Since P is an orthogonal matrix, we have

$$PP^T = P^T P = I$$

Since $\mathcal{S}$ is an orthonormal basis, S is an orthogonal matrix and

$$SS^T = S^T S = I$$

Using that P is the transition matrix from $\mathcal{S}$ to $\mathcal{T}$,

$$\begin{aligned} T &= PS \\ T^T T &= T^T PS = (PS)^T PS \\ T^T T &= S^T P^T PS = S^T S = I \end{aligned}$$

Similarly,

$$\begin{aligned} T &= PS \\ TT^T &= PST^T = PS(PS)^T \\ TT^T &= PSS^T P^T = P^T P = I. \end{aligned}$$

Since $T^T T = TT^T = I$, we have that T is an orthogonal matrix. Since T was formed from the basis vectors of $\mathcal{T}$, we have that $\mathcal{T}$ is an orthonormal basis.

(5) IF MATRIX R_1 GIVES ROTATION THROUGH θ (COUNTERCLOCKWISE), R_2 GIVES ROTATION

THROUGH ϕ THEN WHAT IS THE GEOMETRICAL SIGNIFICANCE OF $R_1 R_2$?

$$\text{Let, } R_1 = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, R_2 = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$

$$\text{Now, } R_1 R_2 = \begin{bmatrix} \cos \theta \cos \phi - \sin \theta \sin \phi & -\cos \theta \sin \phi - \sin \theta \cos \phi \\ \sin \theta \cos \phi + \cos \theta \sin \phi & \cos \theta \cos \phi - \sin \theta \sin \phi \end{bmatrix} = \begin{bmatrix} \cos (\theta + \phi) & -\sin (\theta + \phi) \\ \sin (\theta + \phi) & \cos (\theta + \phi) \end{bmatrix}$$

Thus, we can see that this $R_1 R_2$ is giving us rotation through $(\theta + \phi)$.

Hence, we can say that the geometrical significance of $R_1 R_2$ is that, it's giving us rotation through $(\theta + \phi)$.

(6) IF A, B ARE ORTHOGONAL MATRICES OF SAME ORDER THEN

(i) AB IS ORTHOGONAL

(ii) A^T IS ORTHOGONAL AND

(iii) A^{-1} IS ORTHOGONAL

Using the definition of Orthogonal matrix, i.e. $A^T A = A A^T = I$.

$$\begin{aligned} \text{i) } (AB)^T (AB) &= (AB)(AB)^T \\ B^T A^T A B &= A B B^T A^T \quad \therefore (AB)^T = B^T A^T \\ B^T I B &= A I A^T \quad \therefore A^T A = I \\ B^T B &= A A^T \\ I &= I \end{aligned}$$

$$\begin{aligned} \text{ii) } (A^T)^T A^T &= (A^T)(A^T)^T \quad \therefore (A^T)^T = A \\ A A^T &= A^T A \quad \therefore A^T A = I \\ I &= I \end{aligned}$$

$$\begin{aligned} \text{iii) } (A^{-1})^T A^{-1} &= (A^{-1})(A^{-1})^T \\ (A^T)^T A^T &= (A^T)(A^T)^T \quad \therefore A^{-1} = A^T \\ \text{Now same has part (ii)} \end{aligned}$$

(7) IF $\langle A\underline{u}, A\underline{v} \rangle = \underline{A\underline{u}} \cdot \underline{A\underline{v}}$

WHERE A IS AN $n \times n$ MATRIX,
THEN FOR WHICH MATRIX IT
IS TRUE THAT

$$\langle A\underline{u}, A\underline{v} \rangle = \langle \underline{u}, \underline{v} \rangle$$

FOR EUCLIDEAN INNER PRODUCT

Solution:

$$\langle Au, Av \rangle = (Av)^T(Au) = v^T A^T Au = v^T Iu = v^T u = \langle u, v \rangle$$

This condition holds when A is an orthogonal matrix, i.e. $A^T A = I$.

One can find this condition by equating $\langle Au, Av \rangle$ and $\langle u, v \rangle$, so

$$v^T A^T Au = v^T u \Rightarrow v^T A^T Au - v^T u = 0 \Rightarrow v^T (A^T A - I)u = 0.$$

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Sol:

EIGENVALUES/EIGENVECTORS

(Q) FIND THE BASES FOR THE EIGENSPACES OF

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

We have given $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$

The characteristic equation of the matrix is $\det(A - \lambda I) = 0$

$$\Rightarrow \begin{bmatrix} -\lambda & 0 & -2 \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{bmatrix} = 0$$

$$\Rightarrow -\lambda(2-\lambda)(3-\lambda) + 2(2-\lambda) = 0$$

$$\Rightarrow (\lambda-2)^2(\lambda-1) = 0$$

$$\Rightarrow \lambda = 2, 2 \text{ and } 1$$

Calculating eigenspaces:

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$$E_1 = \{x \in \mathbb{R}^3 \mid (A - I)x = 0\}$$

$$\Rightarrow E_1 = \left\{ x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid \begin{bmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \right\}$$

$$\Rightarrow E_1 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid x = -2z \text{ and } x + y + z = 0 \right\}$$

$$\Rightarrow E_1 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid x = -2z \text{ and } y = z \right\}$$

$$\Rightarrow E_1 = \left\{ \begin{bmatrix} -2z \\ z \\ z \end{bmatrix} \mid z \in \mathbb{R} \right\}$$

$$\Rightarrow E_1 = \left\{ z \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \mid z \in \mathbb{R} \right\}$$

$$\Rightarrow E_1 = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$\text{Bases for the eigenspace } E_1 \text{ is } \left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$$

the eigenspace corresponding to the eigenvalue 2:

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$$E_1 = \{x \in \mathbb{R}^3 | (A - 2I)x = 0\}$$

$$\Rightarrow E_2 = \left\{ x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid \begin{bmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \right\}$$

$$\Rightarrow E_2 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid x + z = 0 \right\}$$

$$\Rightarrow E_1 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid x = -z \right\}$$

$$\Rightarrow E_1 = \left\{ \begin{bmatrix} -z \\ y \\ z \end{bmatrix} \mid y, z \in \mathbb{R} \right\}$$

$$\Rightarrow E_2 = \left\{ z \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \mid y, z \in \mathbb{R} \right\}$$

$$\Rightarrow E_2 = \text{span} \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$\text{Bases for the eigenspace } E_1 \text{ is } \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

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(9) IF k IS A POSITIVE INTEGER,
 λ IS AN EIGENVALUE OF
 MATRIX A , AND x IS A CORRES-
 PONDING EIGENVECTOR, THEN
 PROVE THAT λ^k IS AN EIGEN-
 VALUE OF A^k AND x IS A
 CORRESPONDING EIGENVECTOR.
HINT: PROVE BY MATHEMATICAL
 INDUCTION.

- 1.) $Ax = \lambda x$, given.
- 2.) Inductive hypothesis: $A^k x = \lambda^k x$ for some positive integer k .
- 3.) Operate on the equation of (2) with A : $A^{k+1}x = A(A^k x) = A(\lambda^k x) = \lambda^k (Ax) = \lambda^{k+1}x$.
- 4.) Conclude that $A^k x = \lambda^k x$ for all positive integers k .

(10) PROVE THAT EIGENVECTORS
 OF A CORRESPONDING TO
DISTINCT EIGENVALUES ARE
DISTINCT.

Solution: Assume matrix A has two different eigenvalues corresponding to some eigenvector, such that

$$Ax = \lambda_1 x \quad \& \quad Ax = \lambda_2 x$$

so, $Ax = \lambda_1 x = \lambda_2 x \longrightarrow \lambda_1 x = \lambda_2 x \longrightarrow \lambda_1 x - \lambda_2 x = 0 \longrightarrow (\lambda_1 - \lambda_2)x = 0$
 Hence, this implies $(\lambda_1 - \lambda_2) = 0$ so $\lambda_1 = \lambda_2$.

(11) HOW CAN YOU FIND THE EIGENVALUES OF DIAGONAL, UPPER TRIANGULAR AND LOWER TRIANGULAR MATRICES?

Solution: Eigenvalues of Diagonal & Triangular matrices are the diagonal entries of a matrix.

(12) IF $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

FIND THE EIGENVALUES OF A^{100} .

Solution: Since $Ax = \lambda x$ so, $A^k x = \lambda^k x$. By computing eigenvalues of a matrix A has 1,2,3 so the eigenvalues of a matrix A^{100} is $1^{100}, 2^{100}, 3^{100}$.

(13) IF $\underline{x}$ IS AN EIGENVECTOR OF $\boxed{A}$ CORRESPONDING TO $\boxed{\lambda}$ THEN PROVE THAT $\boxed{\lambda^3}$ IS THE EIGENVALUE OF $\boxed{A^3}$ CORRESPONDING TO EIGENVECTOR $\underline{x}$.

Solution: $Ax = \lambda x \rightarrow A^2x = A(\lambda x) = \lambda(Ax) = \lambda(\lambda x) = \lambda^2x$, similarly for $A^3x = \lambda^3x$.

